STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 7: Gravitation

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Equivalent correct methods should receive credit.

Q1 1 mark

The radius vector sweeps equal areas in equal times.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q2 1 mark

F=Gm1m2/r^2.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q3 1 mark

N m^2 kg-^2.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q4 1 mark

g=GM/R^2.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q5 1 mark

U=-GMm/r.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q6 1 mark

No.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q7 1 mark

Gravity.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q8 1 mark

T^2 proportional to r^3.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q9 2 marks

4.00 x 10- N.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q10 2 marks

490 N.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q11 2 marks

1/sqrt2.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q12 2 marks

1/4.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q13 2 marks

DeltaU=mgh.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q14 2 marks

8.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q15 3 marks

g_h/g=(2/3)^2=4/9.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q16 3 marks

sqrt(2gR)=4.47 x 10^3 m/s.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q17 3 marks

sqrt(GM/r)~7.56 x 10^3 m/s.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q18 3 marks

150 N.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q19 4 marks

GMm/r^2=mv^2/r v=sqrt(GM/r); T=2pir/v=2pisqrt(r^3/GM).

Suggested split: 1 for principle, 1 for equation, 1 for working and 1 for result/explanation. Accept an equivalent correct method.

Q20 4 marks

v~7.56 x 10^3 m/s; E=-GMm/(2r)~-2.86 x 10 J.

Suggested split: 1 for principle, 1 for equation, 1 for working and 1 for result/explanation. Accept an equivalent correct method.